Optimization of a timetable based on graph coloring

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Abstract

We optimize a conference timetable, applying graph coloring. We make use of two graphs. One is for grouping presentations which should be in the same session. In this graph, a vertex represents a speaker, an edge represents a pair of presenters with common coauthors, and a color represents a group. The other is for allocating each group to a session. In this graph, a vertex represents a group, an edge represents a pair of groups that conflict, and a color represents a time slot. The optimal set of groups is obtained as the ground state of a ferromagnetic Potts model on the former graph. An optimized timetable is obtained from the ground state of the antiferromagnetic Potts model on the latter graph.

Keyword: timetable, graph coloring, Potts model, simulated annealing

1 INTRODUCTION

Planning appropriate conference timetables is complicated and difficult. Each speaker may have several coauthors. Speakers who have common coauthors shouldn't give presentation at the same time in different parallel sessions. Sessions with the same topic shouldn't be allocated to the same time slot in different places. The number of presentations in one day is limited, and the number of places is also limited. Taking all the above constraints into consideration, we develop appropriate conference timetables easily. We use two kinds of graphs and apply the idea of graph coloring to optimize a conference timetable.

2 TWO-STEP PROCESS

The approach to solve the conference timetabling problem consists of two steps.

2.1 Grouping presentations

Grouping presentations which should be in the same session is the first step. Presentations in the same group have similar properties, in other words, they have the same coauthors and the same topic. We make a coauthors-network. In this network, each vertex represents a speaker, and linked speakers have common coauthors. Each edge is weighted according to the number of common coauthors. The optimal set of groups is obtained as the ground state of the ferromagnetic Potts model on the network [1]. The vertices which are in the same Potts state (the same color) belong to the same group.

2.2 Allocating each group

After grouping, we allocate each group to a session under two conditions. First, groups with the same topic are not allocated to the same time slot in different places. Second, speakers who have the same coauthors are not allocated to the same time slot in different places. The first condition is prior to the second one. We make a graph to allocate each group to an appropriate session. In the graph, each vertex represents a group, and an edge links the the two groups that should be allocated to different time slots. Coloring this graph gives an appropriate allocation of groups. The coloring problem is regarded as the problem of finding the ground state of the antiferromagnetic Potts model on the network [1]. The color (Potts state) of a vertex represents a time slot. The vertices with the same color are allocated to the same time slot in different places.

3 RESULT

We obtain optimized conference timetables through the two-step process. In both steps, we employ simulated annealing to find the ground state of the ferromagnetic or antiferromagnetic Potts model. Dividing the problem-solving process into two steps and using the same scheme the same scheme in both steps are useful for reducing calculation costs.

References

[1] S. N. Dorogovtsev, A.V. Goltsev, and J. F. F. Mendes, "Critical phenomena in complex networks", Rev. Mod. Phys. vol.80, 1275 (2008).